Experimental Protocol

PDF edition of preserved repository document | docs/experimental_protocol.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Experimental protocol

Freeze datasets and holdout rules before fitting. Record random seed, complete configuration, model identifier/parameter count, dataset version/hash/size, regime, split, hyperparameters, checkpoint, software versions, hardware and runtime. Create unique result directories; never overwrite. Save config, metrics, predictions and summary. Retain failed runs. Use 3-5 seeds where practical, mean and standard deviation and appropriate confidence intervals. Raw graph data must remain available. Ablations must remove operational components rather than only toggling labels. Historical references require exact protocol compatibility; no comparison across unrelated benchmark tasks is valid.

Phase 2 operational rules

Freeze dataset_version, generator_version, manifest content hash and split structures before fitting. The current development dataset tests infrastructure only and may inform infrastructure debugging; do not treat it as an untouched final research test. Allocate new versioned final datasets before neural experiments. Any subsequent validation-informed design or split revision must be recorded.

Train only from a recorded nested subset and regime-specific training view. Regime C receives concepts/limited examples/environment feedback rather than full trace labels; this phase provides its data view, not a learning algorithm. Record procedural knowledge access as an experimental condition. A reference algorithm is not a fair learned baseline and its perfect arithmetic score is not evidence of data efficiency.

Evaluate baseline, structured baseline and Mindscape on identical IID/OOD splits, backbone/parameter count, decoding, epochs and compute where practical. Repeat 3-5 seeds for final studies. Keep accuracy, groundedness, trajectory validity, goal success,

OOD accuracy, runtime and data efficiency separate. N* uses observed sizes only and unreached thresholds remain censored. Retain failures and every prediction file. Run directories record null for unmeasured training/checkpoint fields; future training must populate them. Final test sets must not drive tuning.

First learned development protocol

Before evaluation, record fairness.json: same dataset hash/subset IDs, train/validation and IID/OOD sets, same hidden backbone, seeds, optimizer steps, batch size, learning rate, evaluator and validation-loss selection. Head sizes and supervision count differ. Policy-only access to exact arithmetic transitions and singleton action masks are major confounds. Include unmasked policy runs and raw action validation diagnostics; never infer architecture superiority from the forced masked execution.

The fixed schedule uses budgets 50/100/250; seeds 0/1/2 at 50, seed 0 at larger budgets. These are development results on development_v1, not final untouched thesis results. The protocol and implementation are committed before evaluation. No model/prompt or test-set change is made in response to held-out scores. Match or explicitly vary procedural knowledge/tool access in later experiments. Do not compute DER from the masked algorithm's guaranteed accuracy. Historical reference scores remain unpopulated.

Controlled numerical-claim revision

The revised study is documented in research_protocol.md and milestone_04.md. Prior singleton/oracle-assisted results are exploratory, not current evidence. Primary numerical-claim execution writes model proposals without correction; Regime C uses training-only scalar feedback and positive replay. All four conditions share capacity and the evaluator; remaining procedural/supervision/compute differences are explicit. All reported thresholds are censored; no data-efficiency advantage has been established. Use the research-specific CLI/config files for this study.